

Global E-Commerce Data Stack Production Design for Vendor

Feeds, Inventory, Orders, Returns, and Multi-Granular Sales

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One-Pager Summary

Goal Deliver a global, multi-tenant data platform that ingests vendor catalogs and price lists, warehouse inventory and telemetry, orders and returns, and operational signals, then powers accurate near real time metrics and sales dashboards from SKU to region to enterprise rollups. The platform must provide trustworthy single sources of truth, low-latency operational views, high-throughput historical analytics, and strong governance and privacy.

Key Outcomes

Timely and correct views of inventory, orders, and returns across regions and channels. Business friendly semantic models and metrics for sales, margin, and supply chain KPIs. Low-latency operational APIs for availability and allocations and near real time dashboards. End to end lineage, data quality contracts, and auditability for compliance and trust. Cloud agnostic, cost-aware, and horizontally scalable design with clear SLOs.

Primary Assumptions

Vendors thousands with feeds ranging from hourly to near real time. SKUs millions with regional catalogs and price overrides. Warehouses hundreds across continents with WMS events. Orders sustained two thousand to twenty thousand per second globally with seasonal bursts. Returns rate five to fifteen percent of orders with RMA flows. Analytics users hundreds to thousands, dashboards dozens to hundreds, programmatic consumers many services. Two cloud regions per continent with data residency constraints where required.

Functional Requirements

Ingest vendor feeds for catalog, pricing, availability, and promotions with schema evolution and deduplication. Capture warehouse inventory movements and cycle counts, ASN receipts, pick and pack, and shipment confirmations. Capture orders, payments, fulfillments, cancellations, and returns with RMA states. Provide authoritative product master with hierarchy, attributes, bundles, and channel listings. Expose inventory availability by channel and location and support soft and hard allocations. Provide metrics and dashboards for sales, net revenue, margin, conversion, stockouts, backorders, returns, and fulfillment SLAs at multiple grains. Offer data contracts, lineage, quality rules, anomaly detection, and alerts. Support data exports, partner APIs, and reverse ETL to operational systems. Multi-tenant support by brand, country, or business unit with policy based access.

Non-Functional Requirements

Availability ingestion and operational APIs ninety nine point nine five percent or better. Data freshness for operational inventory target under sixty seconds p95. Orders and returns under five seconds p95 from source event to lakehouse bronze and to serving caches. Dashboard query latency under two seconds p95 for common slices. Operational availability API under one hundred milliseconds p95 from edge. Durability RPO zero for accepted events with multi AZ. RTO under fifteen minutes for warehouse and order domains. Security and compliance SOC two Type two, GDPR, CCPA, and data residency adherence with deletion and export workflows. Cost efficiency with storage tiering, query pruning, and autoscaling.

Architecture Overview

High level view A streaming backbone carries events from OMS, WMS, payment, and storefront systems alongside batch and change data capture from source databases. The lakehouse

implements a medallion pattern with bronze, silver, and gold layers on object storage. A real time serving layer exposes hot operational datasets for inventory and orders. A cloud data warehouse or query engine powers analytics and dashboards through a governed semantic layer. Governance spans catalog, lineage, quality, and policies. CI and CD and IaC keep everything reproducible.

Ingestion

and contracts Public and partner APIs accept vendor feeds in JSON or CSV with schema registry backed validation. CDC from

OMS, WMS, and catalog databases flows through Kafka. File landers accept SFTP and object store drops. All contracts are versioned with Protobuf or JSON schema and checked on ingest with dead letter and quarantine. Streaming and processing Flink or Spark Structured Streaming standardizes events, enforces idempotency, applies currency normalization, and computes incremental aggregates. Business rules such as inventory reservations and allocation soft locks are applied in stream for low latency views. Late data is handled with watermarking and correction jobs. Lakehouse Object storage holds bronze raw facts with append only compaction, silver normalized and conformed entities, and gold curated marts for sales and supply chain. Iceberg or Delta Lake manages table metadata, partition evolution, time travel, and ACID. Compaction, clustering, and Z ordering or sorting maintain performance. Serving and APIs Redis and a key value store back operational APIs for availability, allocations, and order summary views. The analytics warehouse or query engine and the semantic layer power BI and notebooks. A feature store exposes derived features for propensity or forecasting models without duplicating pipelines. Governance A central data catalog tracks schemas, owners, lineage, and usage. Data quality monitors check freshness, volume, nullness, referential integrity, and custom rules. Access controls are policy based with attribute filters for tenant, region, and PII. All changes flow through code review and automated checks.

Technology Stack Edge and auth Cloud CDN, API Gateway, WAF, OAuth two point one or OpenID Connect, mTLS in mesh. Streaming Kafka with Schema Registry and Kafka Connect. Alternative Kinesis for managed. Processing Apache Flink for low latency streams and Spark for heavy batch. Debezium for CDC. Airflow or Dagster for orchestration. Lakehouse S3 or GCS with Iceberg or Delta and Parquet. Compaction jobs and clustering. Warehouse and query Snowflake or BigQuery or ClickHouse or Trino with object storage. Serving Redis Cluster for hot operational views and DynamoDB or Cloud Bigtable or Scylla for durable key value access. Semantic layer and BI dbt and a metrics or semantic layer with Looker or Lightdash or Tableau for dashboards. Observability OpenTelemetry, Prometheus, Grafana, Loki, and DataDog or CloudWatch for platform metrics. Platform Kubernetes for services. Terraform or Pulumi and Helm for IaC. Argo CD and Argo Rollouts for delivery.

Data Model Overview Product product id, tenant id, sku, variant id, title, description, brand, category hierarchy, attributes json, bundle components, status, created at, updated at. Price product id, tenant id, list price, sale price, currency, region, channel, effective start and end, price list id. InventoryEvent tenant id, warehouse id, product id, lot or serial, quantity delta, reason code, source such as receipt or pick or cycle count, event id, occurred at, processed at. InventorySnapshot tenant id, warehouse id, product id, on hand, reserved, available, safety stock, updated at. Order tenant id, order id, user or account id hashed, channel, currency, country, created at, status, totals, tax, shipping, discount, payment status. OrderItem order id, product id, quantity, price, tax, discounts, status, fulfillment group id. Fulfillment fulfillment id, order id, warehouse id, carrier, service level, tracking, shipped at, delivered at, status. Return and RMA return id, order id, product id, reason code, condition, refund amount, received at, disposition, status. Allocation tenant id, warehouse id, product id, reservation id, quantity, state, expires at, created at. MetricFact date and hour and window id, tenant id, grain fields, gross sales, net sales, units, margin, returns, stockouts, backorders, on time, late, cancellations. All tables include audit fields, data source, and version. Keys include tenant id and region to support multi tenancy and residency.

to bronze under five seconds p95 for orders and returns. Vendor feed files processed under fifteen minutes p95 unless near real time feeds are negotiated. Inventory events to serving availability under sixty seconds p95 and under ten seconds for premium warehouses with scanners or IoT. Dashboard tiles from semantic layer under two seconds p95, under one second for most cached top views. Operational availability API under one hundred milliseconds p95 from edge when hitting Redis or KV. Platform availability ninety nine point nine five percent for ingestion and read APIs,

ninety nine point nine percent for writes, and ninety nine point nine percent for dashboards.

Storage Strategy

Bronze Append only raw events in Parquet partitioned by event date and region with schema evolution recording. Data retained for compliance. Silver Conformed entities with deduplication and upserts. Change history maintained with slowly changing dimension type two for product, price, and warehouse reference data. Gold Curated marts by domain. Sales fact tables by day and hour with dimensions for product, channel, region, and campaign. Inventory fact tables by day and minute where necessary for operational KPIs. Wide tables for dashboard speed. Serving Redis holds current availability and allocation tokens. DynamoDB or similar holds authoritative availability by tenant and product and

warehouse with conditional updates for reservations and releases. A read model cache sits in front of the warehouse for the most common dashboard tiles. Analytics and ML Warehouse stores aggregated facts. Feature store materializes lead time forecasts, demand scores, and return risk that are trained from the lakehouse and served to allocation and pricing services.

Scaling Approach Partition Kafka topics by tenant and region and product hash to parallelize processing while preserving per product ordering. Flink job parallelism matches partition counts. Lakehouse tables partition by date and region and bucket by product id to balance files. Redis shards keys by tenant and product and warehouse with hash tags. Serving KV scales horizontally with adaptive capacity and tuneable consistency. Cross region Use regional cells with their own streams and stores and a thin global plane for catalogs, configuration, and identity. Asynchronous replication to comply with data residency and to reduce cross region cost. Global dashboards stitch by querying multiple regions through the semantic layer.

Latency Budgets and SLAs

Key Algorithms and Workflows

Idempotency and dedupe Use event idempotency keys per source. CDC includes transaction id and log position. Streaming jobs track seen sets with short lived state and compaction for long windows. Reservations and allocations Use conditional updates in KV and a short lived token for soft hold with expiry. Confirm holds on payment or fulfillment allocation. Reconcile with WMS events and fix drift with periodic snapshots. Slowly changing dimensions Maintain dimension type two for attributes and price lists with valid from and to. Dashboards can slice by historical attributes that were in effect at the time of sale. Multi currency normalization Convert amounts to a settlement currency at booking time with daily or hourly FX. Keep original currency for reporting. Avoid double conversion on returns. Anomaly detection Rules or ML for price spikes, return abuse, and missing feeds. Alerts dispatched to on-call and vendor managers with runbooks. Data quality Contracts enforce required fields and ranges. System tracks freshness and volume anomalies and blocks promotion to gold when rules fail.

Security, Reliability, and Compliance

Encryption TLS one point three in transit and KMS at rest with optional customer managed keys per tenant. RBAC and ABAC for service and user access. Row level and column level policies mask PII. Audit logs for access and changes. Retention windows per domain with legal hold support. Subject rights for access, export, and deletion. Data residency enforced by region partitioning and access policies. Multi AZ and multi region readiness for critical domains. Disaster recovery with tested restore plans.

Multi Tenancy Tenant id and region are required attributes in every event and row. Logical isolation with tenant scoped topics, tables, and keys. Optional hard isolation for large tenants with dedicated namespaces and VPCs. Per tenant encryption keys, budgets, and quotas. Semantic layer applies row filters and object level permissions so analysts only see their tenants.

CI and CD and Infrastructure as Code Everything as code with Terraform or Pulumi for clusters, networks, topics, storage, and policies. dbt or equivalent manages transformations with tests and documentation. Argo CD and Argo Rollouts deliver services and pipelines with canary and automatic rollback when SLOs

burn. Migrations use online DDL and dual run windows. Contract tests and data tests run in CI and block merges on failure.

APIs and Integrations

Vendor Feed API accepts signed uploads and supports presigned URLs and schema discovery. Partner Webhooks accept catalog and price changes. Inventory API exposes availability and allocation tokens. Orders and Returns API exposes summaries and search. Analytics API or SQL endpoint exposes metrics through the semantic layer. Reverse ETL pushes gold data to CRM, marketing, and ad platforms. Admin APIs manage vendors, warehouses, currencies, and rules. Observability and Operational Practices OpenTelemetry for services and pipelines. Golden signals cover ingestion lag, freshness, row volumes, error rates, compaction health, query latency, and cache hit ratio. Data lineage and usage metrics guide deprecation. SLO aware autoscaling triggers on lag and tail latency. Runbooks for feed delays, schema drift, inventory drift, and warehouse outage. On call rotations by domain with regular game days.

Capacity Planning and Cost Estimate Reference scale five million SKUs, two thousand vendors, one hundred warehouses, and twenty thousand orders per second peak. Kafka nine brokers and two hundred to five hundred partitions for hot topics. Flink parallelism one hundred twenty eight to two hundred fifty six with checkpoints under five seconds. Lakehouse storage cost dominated by Parquet in object storage and compaction jobs. Warehouse compute scales elastically with query concurrency and slot reservations. Redis twelve shards. KV store provisioned for conditioned writes at twenty thousand per second sustained with burst three times. Rough monthly cost across two busy regions between one hundred thousand and three hundred thousand depending on analytics concurrency and retention.

Testing Strategy and Failure Drills

Unit tests for parsing, contracts, currency, and reservation logic. Property tests for idempotency and watermarking. Contract tests with vendors and CDC sources. Load tests for hot product and hot warehouse keys. Soak tests for compaction and small file handling. Chaos drills for broker loss, cold cache, regional isolation, and warehouse outage with reconciliation. Disaster recovery drills verify restore time and data integrity for bronze, silver, and gold layers.

Phased Build Plan

MVP three months single region, vendor feed intake with schema registry, CDC for orders and inventory, lakehouse bronze and silver, basic gold facts for daily sales and inventory, operational availability API over Redis and KV, and a minimal semantic layer for top dashboards. Phase two six to nine months multi region cells, allocation tokens and reconciler, SCD two dimensions and full price lists, anomaly detection, dbt based marts, and standardized reverse ETL. Roll out canary deployments and SLO burn based rollback. Phase three twelve months hard isolation options per tenant, advanced forecasting and ML features, cross region dashboard federation, stronger privacy controls, and cost optimization with storage tiering and materialized views.

Key Choices, Trade Offs, Risks, and Mitigations Lakehouse plus warehouse balance cost and performance while keeping object storage as the system of record. The trade off is operational complexity, mitigated by automation and IaC. Streaming for orders and inventory provides freshness and replay at the cost of stateful processing and checkpoints, mitigated by managed services and strict idempotency. Redis and KV deliver operational latency and conditional correctness but add cache invalidation complexity, mitigated by stream driven invalidations and short TTLs. Data residency and privacy rules increase fragmentation, mitigated by a thin global plane and region aware policies. Vendor feed quality varies widely, mitigated by contracts, quarantine, and runbooks.